



MATHEMATICS

0580/42

Paper 4 (Extended)

October/November 2018

MARK SCHEME

Maximum Mark: 130

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2018 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **8** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	1200	2	M1 for $1962 \div 1.635$
1(a)(ii)	1667.7[0] final answer	2	M1 for $1962 \times (1 - \frac{15}{100})$ oe or B1 for 294.3[0] If 0 scored, SC1 for answer 1020
1(a)(iii)	275	2	M1 for $220 \div \textit{their} (5 - 1)$ soi
1b(i)	165	3	M2 for $\frac{9752 - 3680}{3680} [\times 100]$ oe or $\frac{9752}{3680} \times 100$ oe or M1 for $\frac{9752}{3680}$ or $9752 - 3680$
1b(ii)	51 200	3	M2 for $\frac{74240}{100 + 45} [\times 100]$ oe or M1 for 74 240 associated with 145[%] oe
2(a)	-1.5	3	M1 for $30 + 2x = 9 - 12x$ or $10 + \frac{2}{3}x = 3 - 4x$ M1 for collecting <i>their</i> terms correctly to reach $ax = b$
2(b)	$6ab^2(2b + 3a^2)$ final answer	2	M1 for any correct partial factorisation seen or for correct answer seen
2(c)(i)	$10a^5c^9$ final answer	2	B1 for final answer with $10a^k c^9$ or $10a^5 c^k$ or $ka^5 c^9$
2(c)(ii)	$\frac{8a^6}{c^9}$ or $8a^6 c^{-9}$ final answer	2	B1 for final answer with $\frac{8a^6}{c^k}$ or $\frac{8a^k}{c^9}$ or $\frac{ka^6}{c^9}$ [$k \neq 0$] or for correct answer seen

Question	Answer	Marks	Partial Marks
2(d)	0.5 or $\frac{1}{2}$	3	M1 for $y = \frac{k}{(x+2)^2}$ oe B1 for $k = 50$ or M2 for $2(3+2)^2 = y(8+2)^2$ oe
2(e)	$\frac{7x-x^2}{2(x-2)}$ or $\frac{7x-x^2}{2x-4}$ oe final answer	3	M1 for $5 \times 2 - (x-5)(x-2)$ oe seen M1 for common denominator $2(x-2)$ oe isw
3(a)	Rotation 90 ^[o] clockwise oe Origin oe	3	B1 for each
3(b)(i)	Image at $(-4, -1)$ $(-4, -4)$ $(-2, -4)$	1	
3(b)(ii)	Image at $(3, -1)$ $(5, -1)$ $(3, -4)$	2	B1 for translation by $\begin{pmatrix} 7 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -5 \end{pmatrix}$ or for 3 correct points not joined
3(b)(iii)	Image at $(-2, \frac{1}{2})$ $(-2, 2)$ $(-1, 2)$	3	B2 for 3 correct co-ordinates soi in working or correct size and orientation in wrong position or M1 for $\begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} -4 & -4 & -2 \\ 1 & 4 & 4 \end{pmatrix}$ shown or for statement: enlargement, sf 0.5, (0, 0)
4(a)	$\frac{1}{2} \times 4(x-1) \times (2x+5)[\sin 90] = 30$ oe	M1	
	$8x^2 - 8x + 20x - 20$ or better	B1	correct expansion of brackets
	Completion to $2x^2 + 3x - 20 = 0$	A1	with no errors or omissions seen
4(b)	$(2x-5)(x+4)$	M2	Allow M2 for e.g. $2x(x+4) - 5(x+4)$ then $2x-5 [= 0]$ and $x+4 [= 0]$ M1 for $2x(x+4) - 5(x+4)$ or $x(2x-5) + 4(2x-5)$ or $(2x+a)(x+b) [= 0]$ where $ab = -20$ or $a+2b = 3$ [a, b integers]
	2.5 and -4 cao	B1	

Question	Answer	Marks	Partial Marks
4(c)	11.7 or 11.66 ... or 11.67	3	M2dep for $(4(\text{their } 2.5 - 1))^2 + (2 \times \text{their } 2.5 + 5)^2$ or M1dep for $4(\text{their } 2.5 - 1)$ or $2 \times \text{their } 2.5 + 5$ OR B1 for $\sqrt{20x^2 - 12x + 41}$ and M1dep for substituting $x = \text{their } 2.5$ into $\sqrt{20x^2 - 12x + 41}$ at any stage
5(a)	-3, 17	2	B1 for each
5(b)	Fully correct curve	4	B3 FT for 10 or 11 points or B2 FT for 8 or 9 points or B1 FT for 6 or 7 points
5(c)(i)	Correct ruled tangent for <i>their</i> curve through (0, -17)	1	
5(c)(ii)	(1.7 to 2.2, -1 to 2.5)	1	
5(c)(iii)	[y =] 9x - 17 final answer	3	M2dep for answer [y =] 9x[+] - c OR M1dep for gradient = $\frac{\text{rise}}{\text{run}}$ for <i>their</i> tangent at any point B1 for answer [y =] kx[+] - 17 ($k \neq 0$)
5(d)	y = 3x + 2 ruled correctly and -2.2 ... to -2.1 -0.6 to -0.4 2.6 to 2.8	4	B2 for y = 3x + 2 ruled or B1 for [y =] 3x + 2 soi or y = 3x + k ruled or y = kx + 2 but not y = 2 B2 for all 3 values or B1 for 2 values
6(a)	0.6	1	
6(b)	50.7	3	M2 for $1.2 \times 19 + \frac{1}{2}(19 + 12) \times 1.8$ oe or M1 for method for finding any relevant area
6(c)	17.9	3	M2 for <i>their</i> 50.7 - 1.2×19 [- 10] oe or M1 for 1.2×19 oe seen isw
7(a)	29	1	
7(b)	128	2	FT 180 - 2 (55 - <i>their</i> (a)) M1 for angle OCA or angle OAC = 55 - <i>their</i> (a) soi

Question	Answer	Marks	Partial Marks
7(c)	64	1	FT <i>their</i> (b) $\div 2$
7(d)	116	1	FT $180 - \text{their (c)}$
8(a)	370 or 370.2 to 370.3	2	M1 for $864 \div \text{their time}$
8(b)	991 or 990.5 ...	4	M2 for $864^2 + 928^2 - 2 \times 864 \times 928 \cos 67$ or M1 for correct implicit version A1 for 981100 to 981110
8(c)(i)	313	2	M1 for $180 + 133$ or $360 - 47$
8(c)(ii)	[0]79.5 to [0]79.6 ...	4	M2 for $\frac{928 \times \sin 67}{\text{their } 991}$ or $\frac{864 \times \sin 67}{\text{their } 991}$ oe or M1 for implicit form of either A1 for [angle <i>HGB</i> =] 59.5 to 59.6 ... or [angle <i>HGB</i> =] 53.4 or 53.37 to 53.42 M1 dep for <i>their</i> angle <i>HGB</i> + 20 leading to answer or for $133 - \text{their angle } HGB$ leading to answer
9(a)(i)	42.8 or 42.79 ... nfw	4	M1 for mid-values soi M1 for Σfm where <i>m</i> is any value in interval including boundaries M1 (dep on second M1) for <i>their</i> $\Sigma fm \div 120$
9(a)(ii)	Blocks of height 1.8 4.4 8 2.1 with correct widths	4	B1 for each correct block If B0, SC1 for correct frequency densities seen
9(b)	Valid general comment about distributions	1	e.g. [On average], shoppers spend less time shopping on Wednesday oe
10(a)(i)	$75\,000 \times 60 \times 20$ oe	M1	Allow $\times 1200$ for $\times 60 \times 20$
10(a)(ii)	16.4 or 16.36 ...	3	M2 for $\frac{9 \times 10^7 \times 100}{1000 \times 55 \times 10^4}$ oe or B2 for answer 0.164 or 0.1636 ... or B1 for answer figs 164 or 1636 ... or M1 for figs $9 \div \text{figs } 55$
10(a)(iii)	28.3 or 28.27 to 28.28	3	M2 for $\frac{76}{360} \times 2\pi \times 8.5 + 2 \times 8.5$ oe or M1 for $\frac{76}{360} \times 2\pi \times 8.5$ oe

Question	Answer	Marks	Partial Marks
10(b)(i)	3770 or 3769 to 3770. ...	2	M1 for $\frac{1}{3} \times \pi \times 10^2 \times 36$
10(b)(ii)	3.68 or 3.683 to 3.684 ...	4	M3 for $[r^3 =] \frac{1}{2} \times \text{their (b)(i)} \times \frac{3}{4\pi \times 9}$ oe or M2 for $\frac{4\pi r^3}{3} + \frac{4\pi(2r)^3}{3} = \frac{1}{2} \times \text{their (b)(i)}$ or for $\frac{4\pi r^3}{3} = \frac{1}{1+8} \times \frac{1}{2} \times \text{their (b)(i)}$ or M1 for $\frac{4\pi r^3}{3} + \frac{4\pi(2r)^3}{3}$ or $\frac{1}{2} \times \frac{\pi \times 10^2 \times 36}{3}$ or $\frac{1}{2} \text{ their (b)(i) seen}$ or ratio of vols = 1 : 2 ³ oe seen
11(a)(i)	$\begin{pmatrix} -19 \\ -2 \end{pmatrix}$	2	B1 for answer $\begin{pmatrix} -19 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -2 \end{pmatrix}$ or for $\begin{pmatrix} -9 \\ 6 \end{pmatrix}$ or $\pm \begin{pmatrix} 10 \\ 8 \end{pmatrix}$ seen
11(a)(ii)	3.61 or 3.605 to 3.606	2	M1 for $\sqrt{([-]3)^2 + 2^2}$ oe
11(a)(iii)	$-3m + 5n = 14$ and $2m + 4n = 9$	B1	Accept equivalents
	$[m =] -\frac{1}{2}$ or -0.5 and $[n =] 2\frac{1}{2}$ or 2.5 or $\frac{5}{2}$ with evidence of a correct algebraic method	4	M1 for correctly equating one set of coefficients of <i>their</i> equations or rearranges one of <i>their</i> equations to make <i>m</i> or <i>n</i> the subject e.g. $[m =] \frac{1}{2}(9 - 4n)$ oe M1 for correct method to eliminate one variable for <i>their</i> equations or correctly substitutes <i>their m</i> or <i>their n</i> into the other equation e.g. $-\frac{3(9 - 4n)}{2} + 5n = 14$ oe B1 for one correct answer
11(b)(i)(a)	$-\mathbf{a} + 2\mathbf{c}$	1	
11(b)(i)(b)	$\frac{3}{8}(-\mathbf{a} + 2\mathbf{c})$ or $-\frac{3}{8}\mathbf{a} + \frac{3}{4}\mathbf{c}$ oe	1	FT $\frac{3}{8}(\text{their (b)(i)(a)})$ in simplest form

Question	Answer	Marks	Partial Marks
11(b)(i)(c)	$\frac{1}{2}(5a - 2c)$ or $\frac{5}{2}a - c$ oe	1	
11(b)(i)(d)	$\frac{1}{8}(5a - 2c)$ or $\frac{5}{8}a - \frac{1}{4}c$ oe	2	M1 for a correct unsimplified route
11(b)(ii)	4	1	
12(a)(i)	$\frac{10}{20} \times \frac{9}{19}$ oe	M2	B1 for $\frac{9}{19}$ oe seen
12(a)(ii)	$\frac{62}{95}$ oe	4	M3 for $\frac{6}{20} \times \frac{14}{19} + \frac{10}{20} \times \frac{10}{19} + \frac{4}{20} \times \frac{16}{19}$ oe or $1 - \frac{6}{20} \times \frac{5}{19} - \frac{10}{20} \times \frac{9}{19} - \frac{4}{20} \times \frac{3}{19}$ oe or M2 for the sum of two products of different flavours isw or M1 for one correct product of different flavours isw
12(b)	$\frac{5}{57}$ oe	3	M2 for $N \times \left(\frac{4}{20} \times \frac{3}{19} \times \frac{16}{18} \right) + \frac{4}{20} \times \frac{3}{19} \times \frac{2}{18} \text{ oe}$ or for $3 \left(\frac{4}{20} \times \frac{3}{19} \times \frac{16}{18} \right)$ oe or $1 - \left\{ N \times \left(\frac{4}{20} \times \frac{16}{19} \times \frac{15}{18} \right) + \frac{16}{20} \times \frac{15}{19} \times \frac{14}{18} \right\}$ oe or M1 for $\frac{4}{20} \times \frac{3}{19} \times \frac{k}{18}$ oe seen